Not All Embeddings Are Gaussian: Revisiting Adaptive HNSW Search

Pranav Shenvi, M G Shrivardhan Swaroop, Megha Shah, Kiran Hombal, and Shylaja S S

Abstract—Ada-ef (SIGMOD 2026) sets the search breadth (*ef*) of each HNSW query from a difficulty score that assumes a query's similarities to the database are Gaussian. We test this assumption query by query on 41 public datasets. It fits most text embeddings but departs significantly on nine, six of them learned image, audio and recommendation embeddings. The departure matters less than its size suggests: Ada-ef's score still ranks queries on most of these datasets, and it collapses only on the most extreme ones and when hard queries move away from the bulk. Its practical limit is cost. A long probe and a per-query $d \times d$ variance make it slower than a hindsight-tuned fixed *ef* in 28 of 34 runs. PercEF keeps Ada-ef's score but takes its thresholds from the database's empirical distance percentiles and probes 100 distances. At equal probe length it ranks queries about as well as Ada-ef, yet it is cheaper than the tuned fixed *ef* in 32 of 34 runs and faster in 29. Where the Gaussian fits, Ada-ef buys larger gains in tail recall. We also report pitfalls in evaluating adaptive search, one of which affected our own first evaluation.

Index Terms—approximate nearest neighbor search, HNSW, adaptive search, experimental evaluation, distributional assumptions

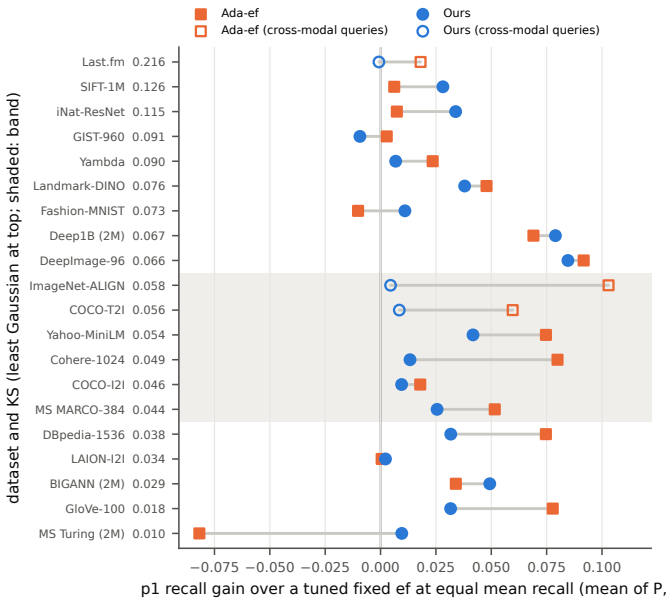


Fig. 1. Tail recall (p1) gained over a fixed *ef* at the same mean recall, for Ada-ef and PercEF, on 20 datasets ordered by KS (higher is less Gaussian). Ada-ef has the larger gain on 10 of the 17 same-modality datasets and PercEF on 7: SIFT-1M, iNat-ResNet, Fashion-MNIST, Deep1B, LAION, BIGANN and MS Turing. KS alone does not separate the two groups. Hollow markers: cross-modal queries.

1 Introduction

Graph indexes such as HNSW [1] answer a nearest-neighbor query by a best-first walk whose breadth is set by one parameter, *ef*. Most deployments pick one *ef* for all queries. That choice is wasteful in both directions: an *ef* large enough for the hardest queries spends work on the easy majority, and an *ef* tuned for the average leaves the hardest queries short of the recall the application asked for. Several recent systems therefore choose the search effort per query, either by learning when to stop [2], [3] or by estimating, before the search, how hard the query will be [4].

Ada-ef [4] takes the second route and needs no trained model.

- It models the similarities between a query and the database as a Gaussian, with mean and variance derived from the database's mean and covariance.
- A short probe of the graph counts how many of the query's nearby points fall in the extreme tail of that Gaussian.
- The count becomes a difficulty score, which maps to an *ef*.

The method is simple and cheap. On the OpenAI and Cohere embeddings of its evaluation, it reaches its target recall with up to 4× lower latency than learned approaches. This paper asks: *does the Gaussian model behind Ada-ef hold on the data it serves, and what decides whether adaptive search pays off when it does not?*

At first sight, the central limit theorem seems to settle the question, since a similarity is a sum over hundreds

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of dimensions. However, the theorem holds only when no small set of dimensions dominates that sum, and whether real embeddings meet this condition has to be checked on the data. The Ada-ef paper supports its assumption on the individual dimensions of GloVe and MS MARCO. It does not report, dataset by dataset, how closely each query's similarities fit the Gaussian, and as far as we can tell no other paper on nearest-neighbor search reports this either: query-hardness measures in the literature [5], [6], [7] either assume such a model or avoid it. We therefore study four questions:

- **Q1.** How often does the Gaussian hold, and does the data type predict it? (§4.2)
- **Q2.** What does Ada-ef lose where the Gaussian departs? (§4.3)
- **Q3.** Does a nonparametric score recover that loss, and what does it give up where the Gaussian holds? (§4.3, §4.4)
- **Q4.** Which component carries each method's benefit and cost: the thresholds, the probe or the *ef* table? (§4.5)

We measure the fit with a Kolmogorov–Smirnov (KS) statistic between each query's similarities and the Gaussian that Ada-ef predicts. We do this on 41 datasets from ann-benchmarks, Big-ANN, VIBE and the Ada-ef paper's own sources (Fig. 2).

The Gaussian departs on nine datasets. It fits most modern text embeddings. The nine where it departs include six learned embeddings:

- CNN features in DeepImage and Deep1B
- DINO and ResNet image models
- an audio model in Yambda
- a factorisation model in Last.fm

The departure hurts Ada-ef's ranking less than its size suggests. On the eight of these datasets whose queries share the data's modality, Ada-ef's score still ranks queries by difficulty better than our alternative in 10 of 16 runs. It breaks down only where the departure is extreme:

- On Fashion-MNIST its rank correlation with the true per-query *ef* is zero.
- On SIFT-1M it is 0.20–0.33, and its *ef* table uses only two values.
- In controlled data where the hard queries move away from the bulk, its rank correlation changes sign.

Ada-ef's cost is the larger problem. Against a hindsight-tuned fixed *ef* at the same mean recall, Ada-ef does less work in 18 of 34 runs, but it is faster in only 6, and in none of the 16 on the non-Gaussian datasets.

PercEF trades ranking for cost. To find which part of the design is responsible, we change Ada-ef in a controlled way. PercEF reads the thresholds from the database's empirical distance percentiles and probes 100 distances instead of 1025. It ranks queries less well than Ada-ef on most datasets, but it is cheaper than the fixed *ef* in 32 of 34 runs, never more than 2.5% more expensive, and faster in 29. To see whether the shorter probe explains this, we ran both scores at probe lengths of 50, 100, 300 and 1025 on all 17 same-modality datasets.

- At equal length, the two rank queries about equally well: PercEF ranks better on 7 to 9 of the 17 datasets.
- PercEF still saves more latency on 14 to 15 of the datasets at every length, and more work at most lengths.
- The latency gap grows with the dimension (Spearman 0.88). Ada-ef computes $q^\top \Sigma q$, a d^2 product, for every query; this alone makes it 39–85% slower than the fixed *ef* on the 1536- and 2048-dimensional datasets, even with a 100-distance probe.

Where the Gaussian holds, Ada-ef's score and *ef* table together buy larger tail gains. Learned early termination (DARTH, LAET) buys the largest tail gains of all, at up to 2.8 times the time of a fixed *ef*.

The overarching lesson is that *a distributional assumption can fail a goodness-of-fit test badly and still give a usable ranking; whether adaptive search pays off is decided mostly by the fixed work it adds to every query, relative to the length of the search.* A second lesson came from our own first evaluation. A reimplemented routine, checked only against itself, gave Ada-ef the wrong thresholds and made the Gaussian look far worse than it is (§4.6).

Contributions.

1) A per-query test of Ada-ef's Gaussian assumption on 41 public datasets. It shows where the assumption holds and that the data type does not predict it (§4.2).
2) An end-to-end re-evaluation of Ada-ef against a hindsight-tuned fixed *ef*:
   - matched on both mean and worst-case recall
   - on 17 same-modality and 3 cross-modal datasets, in two calibration settings
   - against DARTH, LAET and PiP
   - with robustness to the target recall, k, the HNSW degree, the random seed and the metric

   (§4.3–§4.5).
3) PercEF, a nonparametric variant used as a controlled intervention, and an ablation that attributes ranking and cost to the thresholds, the probe and the *ef* table (§4.5).
4) Negative results: the KS test does not predict which score ranks better, and nothing saves work at $k = 10$ or adapts to cross-modal queries. Also seven pitfalls that inflate or distort measured savings of adaptive search (§4.5, §4.6).

2 Related Work

Adaptive search in graph indexes. LAET [2] trains a gradient-boosted model on features gathered early in the search to predict how much longer each query should run. DARTH [3] also terminates early, but targets a declared recall and checks its prediction periodically during the search. Both need a training phase with ground truth and adapt the stopping point while the search runs. Ada-ef [4] instead decides *ef* once, after a short probe, from a statistical model of the query's similarity distribution. It needs only the database mean and covariance and a small calibration

set, which is what makes it cheap and easy to update. PercEF keeps this design. What we examine is the statistical model; these papers do not report its fit dataset by dataset.

Experimental studies of ANN search. Experimental surveys compare graph indexes [8] and ANN methods across many datasets [9], [10], and show that rankings depend on the data. We follow their approach for one design decision, the per-query *ef*, and add something they do not measure: the distributional assumption a method relies on. Distance distributions and their histograms have long been used to characterise how hard a metric space is to search [11]. ADSampling [12] uses the concentration of projected distances to cut distance computations. Queries drawn from a different distribution than the data are known to hurt graph indexes [13], [14]. Our cross-modal results show that they also defeat our empirical score.

Query and dataset hardness. He et al. [5] relate the difficulty of nearest-neighbor search to the relative contrast between nearest and mean distances, derived under a Gaussian-type model of the data. Aumüller and Ceccarello [6] show that local intrinsic dimensionality and related measures vary widely across queries in standard benchmarks and change measured performance. Hubness [15] describes how a few points become nearest neighbors of many queries in high dimensions. Steiner-hardness [7] measures the effort a graph index needs for a query from the index itself. These measures either describe hardness after the fact or need the index. None asks whether the distributional model behind a cheap online score holds, which is the question we answer with a test that runs on the raw vectors.

3 METHODOLOGY

3.1 Preliminaries

Let $X \subset \mathbb{R}^d$ be the database, with vectors normalised to unit length so that cosine similarity and Euclidean distance give the same ranking. For a query q and an integer k, HNSW returns k candidates found by a best-first search whose candidate list holds at most *ef* entries. Recall@k is the share of the true k nearest neighbors among them. We write $\mathrm{ef}^*(q)$ for the smallest *ef* on a grid at which q reaches the target recall τ.

Ada-ef's score. Ada-ef models the distance $\delta(q,x) = 1 - q^\top x$ between q and a random database vector x as

$$\delta(q,x) \sim \mathcal{N}\Big(1 - q^\top \mu,\ q^\top \Sigma q\Big), \tag{1}$$

where μ and Σ are the mean and covariance of X. It places $B = 5$ thresholds $t_1 < \cdots < t_B$ at the $\beta, 2\beta, \ldots, B\beta$ quantiles of this Normal, with $\beta = 10^{-3}$. The thresholds therefore mark the closest tenth of a percent of the database, the next tenth, and so on. An unpruned best-first probe collects the first $L = 1025$ distances the search meets, $P(q)$, and the score is

$$s(q) = \frac{1}{|P(q)|} \sum_{p \in P(q)} w_{b(p)}, \qquad w_b = 100\, e^{-(b-1)}, \tag{2}$$

where $b(p)$ is the first bin whose threshold p falls below; points beyond t_B add nothing. A query surrounded by many unusually close points scores high and is easy; a query whose probe finds few is hard. Offline, Ada-ef scores a calibration set and groups it by rounded score. For each group it stores the smallest *ef* at which the group's average recall reaches τ. Its paper also floors every entry at the weighted average of these values (WAE); the released code does not. We report both variants.

3.2 Testing the Gaussian assumption

Equation (1) can be checked directly. For a query q we compute the similarities $q^\top x$ to a uniform sample of the database, and the KS statistic between their empirical distribution and the Normal implied by (1):

$$\mathrm{KS}(q) = \sup_z \left| \hat{F}_q(z) - \Phi\left(\frac{z - q^\top \mu}{\sqrt{q^\top \Sigma q}} \right) \right|. \tag{3}$$

We report the mean over 200 test queries. We estimate μ and Σ from up to 10^6 database vectors and $\mathrm{KS}(q)$ from 20,000. With 30 queries the estimate was as noisy as the effects we wanted to see, so we use 200 and report a 95% interval. The test needs only the vectors and takes a few seconds per dataset.

3.3 PercEF: a nonparametric control

The thresholds in (2) are the only place where the Gaussian enters Ada-ef's score. To test whether they are what fails, we take them from the data instead.

- Let $c_1, \ldots, c_C$ be centroids of X; $C = 1$ gives the database mean.
- For each centroid, let $t_b^{(c)}$ be the empirical $b\beta$ quantile of the squared distances $\|x - c\|^2$ of the database vectors assigned to it.
- At query time we find the nearest centroid, probe the first $L = 100$ distances, and evaluate (2) with these thresholds and the same weights.

The thresholds thus describe what "unusually close" means in this particular database, whatever the shape of its distance distribution.

Unlike Ada-ef's, these thresholds do not depend on the query. They turn Ada-ef's per-query quantiles into fixed radii measured on the database, and the score counts how many of the probe's points fall within them. Nothing has to be computed per query beyond the probe itself; Ada-ef needs $q^\top \Sigma q$, a d^2 computation. A per-query empirical version, which would estimate each query's own quantiles from a sample, is a natural alternative that we leave to future work.

Calibration.

- We measure $\mathrm{ef}^*(q)$ for every calibration query.
- We fit an isotonic regression [16] of ef^* on the rounded score, letting the fit choose the direction, and clip the result to $[k, 5000]$.
- Optionally, we then multiply the table by one factor so that the mean recall on the calibration queries meets the target (§4.5).

The search does not restart after the probe: it continues the same traversal with the chosen *ef*, so the probe's distance

computations are part of the search, not an addition to it. Our default, fixed before the end-to-end runs, is $C = 1$ with isotonic calibration. We also ran $C \in \{8, 50\}$ and three bucket recipes, and treat them as secondary.

The two scores differ in three ways: their thresholds, the probe length (100 against 1025) and how the *ef* table is built. The rank correlation we report between score and ef* depends only on the first two. We separate them by running each score with the other's probe length (§4.5).

3.4 Implementation

Both methods run inside the same HNSWlib build. Ada-ef runs through its authors' code: we call their scoring, sketch and search routines. We made three changes:

- a loader that reads the database statistics from a file, so that they can be computed in one streaming pass
- a replacement for the Boost normal quantile, which we check against SciPy's
- an estimator that states Eq. (1) in the units of our index: mean $2(1 - q^\top \mu)$ and variance $4\, q^\top \Sigma q$ for squared Euclidean distance on unit vectors, which is exact

On every dataset that fits in memory, we also built Ada-ef's estimator the original way and compared scores on 50 queries. They are identical on 13 of the 17 datasets. On SIFT-1M, BIGANN, DBpedia and COCO-I2I they differ on some queries, by at most 0.39 in score. Our score is computed inside the same traversal, with the same memory prefetching as HNSWlib's own search loop. All searches run on one thread.

4 Experiments

4.1 Setup

Datasets. We benchmark 20 datasets end to end (Table 1):

- the four in the Ada-ef paper that we could obtain: GloVe-100, DeepImage-96, Cohere-1024 and LAION-I2I, the last two as subsets that fit in 62 GB of memory
- a MiniLM-384 stand-in for its MS MARCO set
- standard ann-benchmarks sets [10]
- 2M-vector prefixes of three Big-ANN sets [17]; the benchmark builds its own subsets the same way
- four VIBE sets [18]
- Yambda audio embeddings [19]

Three of the datasets have queries from a different modality than the database (text against images, users against songs).

Protocol. We follow the Ada-ef paper:

- HNSW with $M = 16$ and $ef_{\text{construction}} = 500$
- $k = 1000$ for the MS MARCO, Cohere and LAION sets and $k = 100$ otherwise
- target recall $\tau = 0.95$, and *ef* capped at 5000

Both methods always calibrate on the same set, in two settings: *P*, 200 database vectors as in the Ada-ef paper, and *R*, up to 2000 held-out real queries. Test queries are disjoint from both.

TABLE 1
Datasets, sorted by KS (200 queries). ρ: rank correlation between each score and the true per-query ef^* on the calibration set, settings P / R; the larger magnitude ranks queries better. A dash: the score is constant on the calibration queries.

dataset	source	in Ada-ef paper	corpus	dim	K	test queries	KS	rho Ada-ef P	rho ours P	rho Ada-ef R	rho ours R
MS Turing (2M)	Big-ANN	no	2,000,000	100	100	10000	0.010	-0.81	-0.35	-0.79	-0.40
GloVe-100	ann-benchmarks	yes	1,183,514	100	100	8000	0.018	-0.87	-0.61	-0.88	-0.61
BIGANN (2M)	Big-ANN	no	2,000,000	128	100	8000	0.029	-0.62	-0.74	-0.61	-0.74
LAION-I2I	LAION, shards 0-19	yes (subset)	19,639,187	512	1000	10000	0.034	-0.39	-0.56	-0.40	-0.53
DBpedia-1536	DBpedia, OpenAI embeddings	no	990,000	1536	100	8000	0.038	-0.63	-0.38	-0.66	-0.54
MS MARCO-384	MS MARCO (MiniLM)	stand-in	8,841,823	384	1000	6980	0.044	-0.62	-0.53	-0.64	-0.53
COCO-I2I	ann-benchmarks	no	113,287	512	100	8000	0.046	-0.38	-0.39	-0.40	-0.36
Cohere-1024	Cohere, files 00-04	yes (subset)	9,511,868	1024	1000	1174	0.049	-0.50	-0.40	-0.73	-0.52
Yahoo-MiniLM	VIBE	no	677,305	384	100	700	0.054	-0.58	-0.54	-0.56	-0.68
COCO-T2I	ann-benchmarks	no	113,287	512	100	8000	0.056	-0.39	-0.38	-0.57	–
ImageNet-ALIGN	VIBE	no	1,281,167	640	100	1000	0.058	-0.22	-0.23	-0.24	–
DeepImage-96	ann-benchmarks	yes	9,990,000	96	100	8000	0.066	-0.76	-0.63	-0.75	-0.62
Deep1B (2M)	Big-ANN	no	2,000,000	96	100	8000	0.067	-0.77	-0.64	-0.76	-0.67
Fashion-MNIST	ann-benchmarks	no	60,000	784	100	8000	0.073	+0.05	-0.22	-0.00	-0.29
Landmark-DINO	VIBE	no	760,757	768	100	700	0.076	-0.70	-0.61	-0.64	-0.55
Yambda	Yambda	no	7,709,749	128	100	10000	0.090	-0.38	-0.41	-0.45	-0.43
GIST-960	ann-benchmarks	no	1,000,000	960	100	700	0.091	-0.63	-0.46	-0.62	-0.23
iNat-ResNet	VIBE	no	499,000	2048	100	700	0.115	-0.49	-0.46	-0.36	-0.46
SIFT-1M	ann-benchmarks	no	1,000,000	128	100	8000	0.126	-0.20	-0.33	-0.33	-0.45
Last.fm	ann-benchmarks	no	292,385	65	100	10000	0.216	-0.09	+0.23	-0.00	–

Metrics. Ada-ef's stated aims are to meet a declared target recall while avoiding both under- and over-searching [4]. We use the same recall statistics its evaluation reports:

- mean recall
- the 1st and 5th percentiles of per-query recall (p1, p5), which "capture performance on the hardest queries" and so show under-searching [4, §7.2]

We measure cost in three ways:

- latency per query, as the Ada-ef paper does
- distance computations (DC), including every probe, which the Ada-ef paper does not report and which serve as a hardware-independent measure of cost
- offline time and memory

We count DC as HNSWlib does, as every neighbor-list entry a search examines, visited or not. All methods run in HNSWlib are counted the same way.

The fixed-ef reference. We run HNSW at a grid of fixed *ef* values and, for each method, interpolate the fixed *ef* that reaches the same mean recall. This is the best fixed *ef* anyone could choose in hindsight; no deployment can pick it in advance, so beating it is a demanding test. The Ada-ef paper instead sweeps the fixed *ef* upward until mean recall reaches 0.99 and compares the curves, so it does not match the two at equal mean recall. We interpolate in $\log(1-\text{recall})$ against $\log$ cost. Interpolating linearly overstates the fixed *ef*'s cost by 11% on average in a leave-one-out test on our grids (§4.6), which would flatter every adaptive method.

4.2 How often does the assumption hold?

Figure 2 shows KS for all 41 datasets. Most lie below 0.044. They are mostly text and text-image embeddings (OpenAI, Cohere, MiniLM, Nomic, CLIP and others), but word vectors, LLM attention keys and SIFT-1B are among them too. Nine lie at or above 0.066: SIFT-1M, GIST, Fashion-MNIST, DeepImage, Deep1B, iNaturalist ResNet, Landmark DINO, Yambda and Last.fm.

We fixed the band 0.044–0.066 from our first six datasets and use it throughout to group the results. It is a grouping, not a dividing line between the two methods (§4.5).

Datasets built from the same kind of data land on different sides: SIFT-1M at 0.126 and SIFT-1B at 0.029, Fashion-MNIST at 0.073 and MNIST at 0.037. So the assumption has to be checked on the data itself.

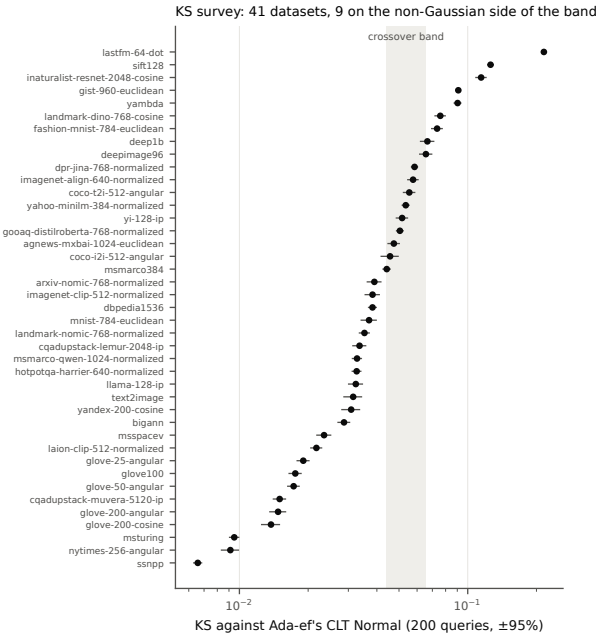


Fig. 2. KS between each query's similarities and Ada-ef's Gaussian, mean over 200 queries with 95% interval, for 41 public datasets. Shaded: the band 0.044–0.066 we use to group results. GloVe-200 appears twice, from ann-benchmarks and from VIBE.

4.3 Comparison on non-Gaussian datasets

Table 2 summarises every run against the fixed-*ef* reference, and Fig. 3 shows where each method sits relative to the fixed-*ef* curve.

Ranking. On the eight same-modality datasets above the band (16 runs), Ada-ef's score has the larger rank correlation with ef^* in 10. On DeepImage, Deep1B and Landmark-DINO it ranks queries well, with $|\rho|$ = 0.64–0.77 against PercEF's 0.55–0.67, even though KS is 0.066–0.076 there. Its score breaks down where the departure is extreme:

- On Fashion-MNIST it is uncorrelated with ef^* ($\rho = +0.05$ and 0.00) and gives every calibration point the same *ef* in setting P.
- On SIFT-1M its correlation is 0.20 and 0.33, against PercEF's 0.33 and 0.45, and its table uses only two distinct *ef* values.

GIST (KS 0.091) is the opposite case: Ada-ef ranks clearly better there (0.62 against 0.46 and 0.23).

Cost. Cost separates the methods more clearly than ranking does.

- PercEF is cheaper than the fixed *ef* in 14 of the 16 runs. The two exceptions are small: DeepImage P costs 2.5% more and GIST R 0.8% more.
- Ada-ef is cheaper in 9 of the 16 runs and faster in none. Its probe makes it 4–93% slower than the fixed *ef* at the same recall.
- Ada-ef's worst run is Fashion-MNIST, at 24.8% more work. There most queries reach the target at the smallest *ef*, so a long probe is pure overhead.

TABLE 2
Runs in which each method is cheaper than the fixed *ef* at the same mean recall (DC), faster (latency, every run timed), and has at least its p1 tail recall; and runs in which it is faster than the fixed *ef* with the same p1 (p1-matched). Same modality unless marked. Worst: the largest extra cost (negative: more work than the fixed *ef*).

		same mean recall				p1-matched
Datasets	Method	Cheaper	Worst	Faster	p1 ≥	Faster
All (34 runs)	Ours	32/34	−2.5%	29/34	31/34	32/34
	Ada-ef	18/34	−24.8%	6/34	29/34	13/34
	Ada-ef WAE	24/34	−24.8%	8/34	29/34	16/34
KS ≥ 0.066 (16)	Ours	14/16	−2.5%	11/16	13/16	14/16
	Ada-ef	9/16	−24.8%	0/16	13/16	4/16
Band (8)	Ours	8/8	+3.4%	8/8	8/8	8/8
	Ada-ef	3/8	−4.5%	3/8	8/8	6/8
KS < 0.044 (10)	Ours	10/10	+2.4%	10/10	10/10	10/10
	Ada-ef	6/10	−15.2%	3/10	8/10	3/10
	Ada-ef WAE	9/10	−4.4%	5/10	7/10	4/10
Cross-modal (6)	Ours	2/6	−1.0%	3/6	5/6	5/6
	Ada-ef	1/6	−20.2%	0/6	6/6	1/6

Tail recall. Both methods usually keep or raise the tail: p1 is at or above the fixed *ef*'s in 13 of 16 runs for each.

- Ada-ef's p1 gains are larger on DeepImage (0.090 and 0.094 against 0.081 and 0.088), Landmark-DINO and Yambda.
- PercEF's are larger on SIFT-1M (0.028–0.029 against 0.004–0.008), iNaturalist (0.030–0.038 against 0.004–0.011), Fashion-MNIST and Deep1B.

Matched on p1. Matching the fixed *ef* on mean recall credits each method with the work it saves but not with the tail it buys. Matched instead on p1, that is, against the cheapest fixed *ef* that serves the worst 1% of queries as well:

- PercEF is faster in 14 of the 16 runs, by a median of 16%.
- Ada-ef is faster in 4.

4.4 Comparison on near-Gaussian datasets

In and below the band, Ada-ef's model fits and its score shows it. It ranks queries better than PercEF in 12 of the 18 runs, by the widest margins on:

- MS Turing: 0.79–0.81 against 0.35–0.40
- GloVe: 0.87–0.88 against 0.61
- DBpedia, MS MARCO and Cohere

Its tail gains are larger too: p1 rises by 0.071–0.084 on GloVe, 0.048–0.056 on MS MARCO and 0.060–0.100 on Cohere, against PercEF's 0.029–0.034, 0.025–0.026 and 0.009–0.017.

With its WAE floor, Ada-ef is also much cheaper than the fixed *ef* on GloVe (26–27%) and MS Turing (6–11%), more than PercEF's 5–6% and 2–3%. On GloVe it is faster in wall-clock time by 21%.

PercEF, in turn, is cheaper and faster than the fixed *ef* in all 18 runs, never lowers its tail, and is faster than its own p1-matched fixed *ef* in all 18. Ada-ef as shipped is cheaper in 9 and faster in 6.

The near-Gaussian side is therefore a real trade-off. Ada-ef is the better choice where searches are long and the worst-served queries matter most; PercEF is the better choice where a predictable cost does.

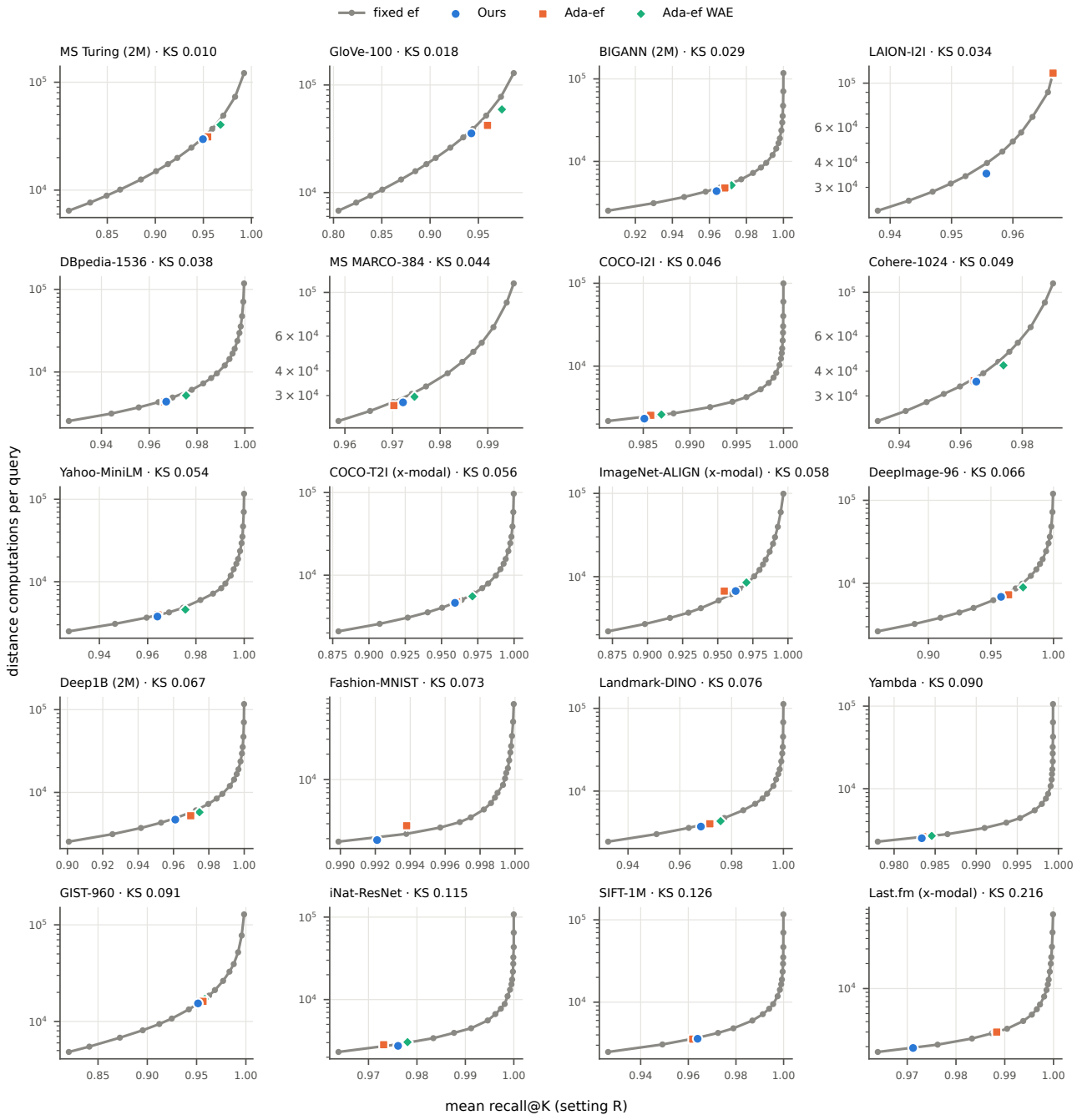


Fig. 3. Distance computations per query against mean recall, setting R. Grey: HNSW with fixed *ef*. Points below the curve reach the same recall with less work. Panels are ordered by KS, from near-Gaussian (top left) to non-Gaussian (bottom right).

Two datasets on this side behave differently:

- **BIGANN** (SIFT-1B, KS 0.029) and **LAION**: our score ranks queries better (0.74 against 0.61–0.62 on BIGANN; 0.53–0.56 against 0.39–0.40 on LAION) and saves the most work of any method (7–8% and 11%).
- **MS Turing** (KS 0.010): Ada-ef ranks best, yet its tail is worse than the fixed *ef*'s (p1 -0.086 and -0.078). Its group-average table under-provisions some score groups; the WAE floor narrows the loss to -0.067 and -0.006.

4.5 Analysis

Does KS predict the better score? We had set the band from our first six datasets. Before running each of the other fourteen, we wrote down which score would rank its queries better. Against the corrected Ada-ef (§4.6), these predictions held on only 4 of the 9 datasets for which we made a definite prediction.

Across the 17 same-modality datasets, KS correlates with Ada-ef's rank correlation at only -0.43 (Spearman). Datasets with moderate KS divide in both directions:

- DeepImage, Deep1B, Landmark-DINO and GIST,

where Ada-ef ranks well
- BIGANN and LAION, below the band, where PercEF ranks better

KS therefore tells whether Ada-ef's model is accurate, not whether its ranking is usable. It is not a rule for choosing between the two scores. Nor does KS single out the collapses: SIFT-1M and Fashion-MNIST are non-Gaussian, but so are DeepImage and GIST, where Ada-ef ranks well. A cheap check helps in practice. Ada-ef's calibration already scores a few hundred points, and when its table maps them to one or two *ef* values (SIFT-1M, Fashion-MNIST, and Yambda in setting P), the score cannot separate queries at run time, whatever its correlation.

Which component carries the benefit? Ada-ef's score has three parts: Gaussian thresholds, a 1025-distance probe and a group-average *ef* table. PercEF changes the first two, so we ran each score with the other's probe length (Table 3).

- **Ranking.** At equal probe length, the empirical thresholds rank queries better on SIFT-1M and BIGANN, by 0.07 to 0.11. The Gaussian thresholds rank better on Deep1B, DeepImage and MS Turing at both lengths, and on GloVe at 1025 distances. The two tie on GloVe at 100. So the thresholds matter, but in Ada-ef's favour on most of these datasets.
- **Probe length and the near-Gaussian edge.** Ada-ef's edge on GloVe and MS Turing comes from its long probe. Cut to 100 distances, its rank correlation falls from 0.88 to 0.62 on GloVe and from 0.79 to 0.43 on MS Turing.
- **Probe length and cost.** The probe explains part of the cost difference, not all of it. We ran both scores at 50, 100, 300 and 1025 distances on all 17 same-modality datasets (Table 4).
  - At equal length, PercEF saves more latency than Ada-ef on 14–15 datasets and more work on 10–16. Its rank correlation is higher on only 7–9.
  - Each method at its own best probe length: PercEF is faster than the fixed *ef* on 15 of the 17 datasets, Ada-ef on 4, and Ada-ef with the WAE floor on 8.
  - The latency gap at equal length correlates with the dimension at 0.88 (Spearman, 16 datasets). Ada-ef computes $q^\top \Sigma q$ for every query, a d^2 product: with a 100-distance probe it is still 39% slower than the fixed *ef* on DBpedia ($d = 1536$) and 85% on iNaturalist ($d = 2048$), but within 3% on all six datasets with $d \leq 128$.

  So on low-dimensional data, Ada-ef with a short probe costs about what PercEF does. On high-dimensional data, its per-query variance does not.
- **Tail.** Better ranking alone does not buy Ada-ef's tail. With 1025 distances, PercEF's p1 gain on GloVe stays at 0.028 against Ada-ef's 0.082. With Ada-ef's own table (each score group gets the smallest *ef* at which the group's average recall reaches the target), PercEF's p1 gain on GloVe is 0.031. Ada-ef's tail on these datasets needs its score and its table together. On the non-Gaussian sets the same table lowers PercEF's tail slightly (DeepImage 0.088 to 0.073), so we keep PercEF's own calibration.

TABLE 3
Rank correlation $|\rho|$ between each score and ef^* when both scores use each probe length L (setting R). Bold: the better score at that probe length.

		Ada-ef		PercEF	
Dataset	KS	L=1025	L=100	L=100	L=1025
SIFT-1M	0.126	0.33	0.35	**0.45**	**0.40**
Deep1B	0.067	**0.76**	**0.74**	0.67	0.63
DeepImage	0.066	**0.75**	**0.72**	0.62	0.59
SIFT-1B	0.029	0.61	0.65	**0.74**	**0.72**
GloVe	0.018	**0.88**	0.62	0.61	0.76
MS Turing	0.010	**0.79**	**0.43**	0.40	0.58

TABLE 4
Probe-length sweep, both scores at each probe length L, 17 same-modality datasets, setting R: datasets on which PercEF is ahead of Ada-ef (as shipped) at the same L. Savings are against the tuned fixed *ef* at the same mean recall; a dataset counts where both have a value. Bottom: each method at its own best L.

L	Higher $\|\rho\|$	More work saved	More time saved	Larger p1 gain
50	8/17	16/17	15/17	10/17
100	9/17	12/16	14/16	8/16
300	8/17	12/16	14/16	7/16
1025	7/17	10/17	15/17	6/17
best L each	–	11/17	15/17	–

Latency and offline cost. Ada-ef's wall-clock cost depends on how long the searches are. Its probe is a fixed 1025 distances.

- **Long searches.** On searches of several milliseconds the probe is a small share, and there Ada-ef can be faster than the fixed *ef* at the same recall: the WAE variant by 21% on GloVe and 9% on MS Turing R; the shipped variant by 6–12% on Cohere.
- **Short searches.** The probe dominates. Ada-ef is 4–93% slower than the fixed *ef* on the non-Gaussian datasets, and 32–115% slower on Last.fm, where a whole search takes 0.13 ms.
- **PercEF.** At the same mean recall, PercEF is faster than the fixed *ef* in 29 of 34 runs: all 18 in or below the band, by 1% to 16%, and 11 of the 16 non-Gaussian runs. In the other 5 it is slower by at most 2.5%, except on DeepImage (8.3%). Against Ada-ef directly, PercEF is faster in 31 of 34 runs.

Why time saved lags work saved. Not every distance computation costs the same. Under a plain fixed *ef*, compare queries with the same distance count (Table S1):

- the queries PercEF classifies as hard (top 20% of its *ef*) take a median 8% more time
- the queries it calls easy take 11% less

This is probably because hard queries wander through less cached parts of the graph. PercEF moves work from easy queries to hard ones, so it saves cheap distance computations and spends dearer ones. Its time per distance computation is a median 1% above the fixed *ef*'s: 0.3% above the band and 2.5% in or below it.

Offline cost. Both methods need seconds for their statistics. Calibration time is dominated by two steps:

- measuring ef^* for PercEF, up to 257 s on MS Turing
- Ada-ef's per-group search, 183 s on MS Turing and up to 370 s on GloVe

Ada-ef keeps a $d \times d$ covariance, up to 9.4 MB at $d = 1536$. Ours keeps five thresholds, the centroid and the table: a few kilobytes.

Other targets and k. All results so far use the Ada-ef paper's protocol ($\tau = 0.95$, $k = 100$ or 1000). We repeated the comparison at $\tau = 0.90$ and 0.99 and at $k = 10$, in both settings (Table 5), on six datasets:

- four above the band: DeepImage, Deep1B, SIFT-1M and Landmark-DINO
- two below it: GloVe and MS Turing

At both other targets, the picture of §4.3 holds:

- Above the band, PercEF does less work than the tuned fixed *ef* in 7 and 6 of the 8 runs, never more than 2.2% extra. It never lowers p1 and is faster than the p1-matched fixed *ef* in all 8, by a median of 18–19%.
- Ada-ef does less work in 5 and 8 of the 8 runs, but it is faster than the fixed *ef* at the same recall in at most 1.
- Below the band, PercEF wins on every count in all 4 runs at both targets.
- At $\tau = 0.99$, Ada-ef with the WAE floor is cheaper than the fixed *ef* in all 8 runs above the band.

Weakness 1: PercEF's mean recall lands further from the target. Our table maps a score to the average ef^* of similar calibration queries and never checks the mean recall that results; Ada-ef's table is built to meet the target. Every comparison above is made at the method's own recall, so the counts are not affected. The gap closes with one offline step: we multiply the whole table by a single factor, the smallest whose mean recall on the calibration queries reaches τ. It is found by bisection, and nothing changes per query. With it, PercEF's mean recall lands within 0.01 of the target:

- above the band: in 7 of the 8 runs at $\tau = 0.95$ (Ada-ef: 2), and in all 8 at $\tau = 0.99$ (Ada-ef: 8)
- below the band: in 3 of the 4 runs at each target

Its advantage over the fixed *ef* survives. Above the band it still does less work in 6 and 7 of the 8 runs, at most 4.5% extra, and is faster than the p1-matched fixed *ef* in all 8, by a median of 17–19%.

Weakness 2: at $k = 10$ neither method saves work.

- PercEF needs up to 19% more than the tuned fixed *ef* above the band, cheaper in 0 of 8 runs.
- Ada-ef needs up to 69% more, and is cheaper in 1 of 12 runs.

The probe is not the cause: with a 30-distance probe, PercEF is still cheaper in none of the 8 runs. A likely reason is that recall moves in steps of 0.1 per query, so each query's ef^* is too coarse for any score to predict. We leave small k open. For the same reason the p1-matched comparison is not meaningful at $k = 10$.

TABLE 5
Robustness sweep, six datasets, settings P and R. Counts against the tuned fixed *ef*: less work at the same mean recall, p1 at least the fixed *ef*'s, and faster than the p1-matched fixed *ef* (median time saving). Above: 4 datasets above the band (8 runs); below: 2 below it (4).

τ, k	Data	Method	Less work	p1 $\geq$	p1-matched faster
0.90, 100	above	Ours	7/8	8/8	8/8 (18%)
		Ada-ef	5/8	6/8	3/8
	below	Ours	4/4	4/4	4/4 (11%)
		Ada-ef	0/4	2/4	2/4
0.95, 100	above	Ours	7/8	8/8	8/8 (24%)
		Ada-ef	7/8	8/8	4/8
	below	Ours	4/4	4/4	4/4 (10%)
		Ada-ef	2/4	2/4	2/4
0.99, 100	above	Ours	6/8	8/8	8/8 (19%)
		Ada-ef	8/8	6/8	4/8
	below	Ours	4/4	4/4	4/4 (11%)
		Ada-ef	3/4	2/4	2/4
0.95, 10	above	Ours	0/8	8/8	–
		Ada-ef	0/8	6/8	–
	below	Ours	2/4	3/4	–
		Ada-ef	1/4	3/4	–

Index degree, seeds and metric.

- **HNSW degree.** At $M = 8$ and $M = 32$ on SIFT-1M, Deep1B, GloVe and MS Turing, PercEF is cheaper than the fixed *ef* in 14 of 16 runs and Ada-ef in 11. Ada-ef is faster in 4 of 16 and PercEF in 8. Ada-ef keeps its edge in work on GloVe, and PercEF on SIFT-1M and MS Turing, as at $M = 16$.
- **Seeds.** With three random seeds for the calibration and test split on SIFT-1M, DeepImage, GloVe and MS Turing, the ordering of the two methods is the same in every seed. PercEF's saving keeps its sign except on DeepImage, where it ranges from -6.1% to $+2.7\%$.
- **Metric.** On raw, unnormalised vectors with Euclidean distance (SIFT-1M, GIST, Fashion-MNIST, BIGANN, Deep1B, MS Turing), PercEF is cheaper than the fixed *ef* in 11 of 12 runs, by up to 7.7% (BIGANN). Ada-ef's estimator assumes unit vectors, so it does not apply.

Learned methods and PiP. DARTH [3] stops each search when a gradient-boosted model, fed with features of the search so far, predicts that the target recall has been reached. LAET [2] predicts once, after a fixed number of distance computations, how many the query needs in total.

We ran both through DARTH's authors' FAISS fork, which contains LAET's search. It is unchanged apart from a build path and a data loader. We ran on seven datasets: GloVe, MS Turing and DBpedia below the band, and DeepImage, Deep1B, Landmark-DINO and SIFT-1M above it.

- They use the same test queries and ground truth, $M = 16$, $ef_{\mathrm{construction}} = 500$ and one thread.
- They are trained on 1,500 calibration queries (DARTH's paper uses 10,000) and tuned on the other 500: DARTH's prediction intervals as in its authors' scripts, and LAET's prediction point and multiplier so that validation recall meets the target.

- Each method is compared with a fixed *ef* of its own library, because the two HNSW implementations build different graphs, time queries differently and count work differently. FAISS counts only the distances it computes: at the same recall its fixed *ef* reports 0.59 to 0.71 times HNSWlib's count on these datasets (Table 6).

DARTH has the best p1 gain on four of the seven datasets, Ada-ef on two and LAET on one. Both learned methods pay for their tail gains in time:

- They are slower than their fixed *ef* on every dataset but one (LAET on GloVe, 1.3% faster), by up to 2.8 times where a search takes about a millisecond and the model calls dominate.
- They do less work than the fixed *ef* only on the two long-search datasets, GloVe and MS Turing, by 7–18%, and 3–20% more on the other five.
- Each needs 2 to 10 minutes of training and tuning per dataset and gigabytes of search traces.

On seven more datasets (BIGANN, COCO-I2I, Fashion-MNIST, GIST, iNaturalist, Yahoo-MiniLM and Yambda) we ran DARTH alone. It is 53–85% slower than its fixed *ef* on the three where the two can be matched. On the other four, its mean recall stays below that of FAISS's smallest fixed *ef*.

Corrected, Ada-ef also does less work than the fixed *ef* on all seven datasets in Table 6, and more savings than PercEF on four of them. But it is faster only on GloVe and MS Turing. PercEF is faster than its fixed *ef* on five of the seven, and at most 8.3% slower on the others.

PiP (patience in proximity), the early-stopping rule that Ada-ef's code base ships, behaves differently:

- At its paper's setting it falls well short of the target (mean recall 0.79–0.81 on SIFT-1M and Deep1B).
- Tuned on our calibration queries, it saves between -1.4% and $+1.7\%$ of the work.
- Its implementation copies the result list at every step, so we compare it on work only.

Cross-modal queries. The cross-modal datasets are Last.fm (user vectors against song vectors), COCO text-to-image and ImageNet ALIGN text-to-image. On them, our score is constant on real queries: every query gets the same *ef*. Real queries lie far from every database vector, so no probe distance reaches the region the empirical thresholds describe. PercEF then behaves as a fixed *ef* at about the right value, between 1.0% more and 0.6% less work.

Ada-ef's query-dependent thresholds still separate these queries, and it buys large tail gains (p1 $+0.084$ on ImageNet ALIGN R, $+0.085$ on COCO-T2I R). But it pays heavily for them: up to 20% more work than the fixed *ef* (ImageNet ALIGN R), and 25–115% more time.

Other graph indexes. We also wrapped FAISS HNSW and NSG. Neither library can continue a search after the probe, so every adaptive query pays for the probe and then a fresh search. Both scores then come out 5–72% more expensive than the fixed *ef*, which says more about the wrapper than about the methods. We report no savings for these indexes. The rank correlations transfer: PercEF's is 0.52–0.76 on Deep1B, GloVe and MS Turing under both libraries.

TABLE 6
Learned methods (DARTH, LAET; FAISS), Ada-ef (as shipped) and PercEF (HNSWlib) against a tuned fixed *ef* of their own library at the same mean recall (setting R). Positive savings: cheaper or faster than the fixed *ef*. Bold: best per dataset. GloVe, MS Turing, DBpedia: below the band; the rest above it.

Dataset	Method	p1 gain	DC saving	Latency saving
GloVe	DARTH	+0.080	+15.5%	−6.3%
	LAET	+0.041	+17.6%	+1.3%
	Ada-ef	**+0.084**	**+21.2%**	**+10.9%**
	PercEF	+0.029	+5.9%	+2.2%
MS Turing	DARTH	+0.044	+6.7%	−41.9%
	LAET	**+0.051**	**+16.4%**	−16.2%
	Ada-ef	−0.078	+6.7%	**+2.2%**
	PercEF	+0.013	+2.4%	+1.0%
DBpedia	DARTH	+0.067	−20.3%	−62.9%
	LAET	+0.057	−8.4%	−60.0%
	Ada-ef	**+0.077**	+4.7%	−57.7%
	PercEF	+0.034	**+5.3%**	**+1.7%**
DeepImage	DARTH	**+0.119**	−10.4%	−112%
	LAET	+0.097	−5.0%	−115%
	Ada-ef	+0.094	**+5.5%**	−13.3%
	PercEF	+0.088	+0.5%	**−8.3%**
Deep1B	DARTH	**+0.098**	−9.4%	−92.1%
	LAET	+0.090	−10.4%	−55.8%
	Ada-ef	+0.075	**+8.3%**	−33.4%
	PercEF	+0.081	+3.7%	**−2.5%**
Landmark-DINO	DARTH	**+0.078**	−12.0%	−179%
	LAET	+0.034	−11.5%	−157%
	Ada-ef	+0.051	+2.8%	−37.8%
	PercEF	+0.039	**+4.0%**	**+2.2%**
SIFT-1M	DARTH	**+0.048**	−2.8%	−126%
	LAET	+0.031	−8.1%	−78.2%
	Ada-ef	+0.004	+0.7%	−41.0%
	PercEF	+0.028	**+2.7%**	**+0.4%**

Ada-ef's score, computed there from the 100 nearest probe distances, cannot be compared fairly.

Controlled data. Synthetic data with growing anisotropy (B) moves the ranking advantage of the empirical score steadily, from -0.45 at KS 0.018 to $+0.20$ at KS 0.127 (Fig. S1 in the supplementary material). Cluster separation (D) does not move it in one direction.

The clearest case is SIFT-1M with hard queries shifted away from the bulk (A):

- As the shift grows, Ada-ef's rank correlation falls from -0.30 through zero to $+0.19$, the wrong sign.
- PercEF's rises from -0.48 to -0.83.
- At Ada-ef's recall, PercEF needs 3–8% less work at the three largest shifts.

On the Gaussian synthetic data, Ada-ef ranks far better (-0.82 against -0.37) and needs 10.6% less work.

4.6 Pitfalls in evaluating adaptive search

Several of our own intermediate results did not survive closer measurement, and the reasons apply to any adaptive-*ef* method.

- **Check a reimplementation against a reference, not against itself.** To build Ada-ef without Boost, we first replaced its normal quantile with a hand-written approximation. Its tail branch returned the wrong sign: $+3.02$ instead of -3.09 at $p = 10^{-3}$. Ada-ef's

thresholds then sat three standard deviations above the mean distance instead of below it. Our check had compared streamed with in-memory statistics through the same function, so both sides agreed. The symptom, a score that was constant on SIFT-1M and Yambda, looked like the effect we were studying. A test against SciPy's quantiles found the error, and every Ada-ef number in this paper comes from the corrected code. With the wrong thresholds, Ada-ef looked much worse on the non-Gaussian datasets than it is.

- **Interpolate the fixed-*ef* reference in the right space.** Interpolating linearly between grid points credited every method with 3–4% of savings it did not have. We found it because two methods that assign one *ef* to every query were credited with savings over a fixed *ef*.
- **Time in rounds.** Timing each method once, in sequence, made latency vary by 24% between identical runs. We time in three rounds with the method order rotated and take per-query medians.
- **Use enough queries for KS.** KS from 30 queries was as noisy as the width of the band.
- **Count the probe.** Ada-ef's fixed probe is a large share of a short search.
- **Check that the target is reachable.** About 2% of LAION queries cannot reach the target at any *ef* up to the cap, which limits every method's p1 there. We report LAION's tail at p5.
- **Match on both mean and tail recall.** Matching on mean recall alone hides what a method buys in the tail, and matching on p1 alone hides what it costs on average. An adaptive method moves work between queries, so both are needed.

5 Lessons

Test the assumption per dataset, not per data type. The Gaussian holds for most modern text embeddings and fails for several learned image, audio and recommendation embeddings, and closely related datasets fall on opposite sides (SIFT-1M and SIFT-1B, MNIST and Fashion-MNIST). A KS test on a few hundred queries, which needs no index, tells which side a dataset is on.

A failed fit is not a failed method. Ada-ef's score still ranks queries well on most datasets where its Gaussian fails a KS test. It collapses only at the extreme, and in controlled data where the hard queries leave the bulk. Whether a model is accurate and whether it is useful are separate questions, and both need measuring.

Fixed per-query overhead decides the outcome. Ada-ef's ranking edge comes largely from a 1025-distance probe, and every query also pays for a d^2 variance. Both are affordable on searches of several milliseconds, and they dominate shorter searches and high dimensions. A shorter probe with fixed empirical thresholds gives up some ranking and is cheaper and faster than a tuned fixed *ef* almost everywhere. Learned early termination shows the same trade at a larger scale.

Match the baseline on both mean and tail recall. At equal mean recall, an adaptive method is credited with the work it saves but not with the tail it buys; at equal tail recall, the reverse. Reporting both, against a fixed *ef* tuned with hindsight, separates methods that save work from methods that move it to the hardest queries.

Validate reimplementations against an independent reference. Our first evaluation of Ada-ef rested on a reimplemented routine that agreed with itself and was wrong. Reproducing a method is part of evaluating it.

6 Conclusion

We asked whether the Gaussian model behind Ada-ef holds on the data it serves. On 41 datasets it fits most modern text embeddings and departs on nine, six of them learned embeddings, and the data type does not tell which.

Where it departs, Ada-ef's score usually still ranks queries. It fails only at the extreme (SIFT-1M, Fashion-MNIST) and when the hard queries move away from the bulk. Its practical limit is the cost of its probe: it is slower than a hindsight-tuned fixed *ef* in 28 of 34 runs.

Replacing its thresholds with the data's own percentiles and shortening the probe gives a method that ranks less well but is cheaper than the tuned fixed *ef* in 32 of 34 runs and faster in 29. It holds across target recalls, index degrees, seeds and metrics. Where the Gaussian holds, Ada-ef's score and table together buy larger tail gains. Neither method saves work at $k = 10$, and our score does not adapt to cross-modal queries.

Code, data pointers and the result files from which every number in this paper is generated are available at https://anonymous.4open.science/r/PercEF/.

Acknowledgments

Generative AI (Claude, Anthropic) was used to write and refactor experiment and analysis code, and to draft and edit text. The authors designed the study, chose and checked every experiment, verified every reported number against the released result files, and take full responsibility for the content.

References

[1] Y. A. Malkov and D. A. Yashunin, "Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 4, pp. 824–836, 2020.

[2] C. Li, M. Zhang, D. G. Andersen, and Y. He, "Improving approximate nearest neighbor search through learned adaptive early termination," in *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data*, 2020, pp. 2539–2554.

[3] M. Chatzakis, Y. Papakonstantinou, and T. Palpanas, "DARTH: Declarative recall through early termination for approximate nearest neighbor search," *Proceedings of the ACM on Management of Data (SIGMOD)*, 2026, arXiv:2505.19001.

[4] C. Zhang and R. J. Miller, "Distribution-aware exploration for adaptive HNSW search," *Proceedings of the ACM on Management of Data (SIGMOD)*, 2026, extended version: arXiv:2512.06636.

[5] J. He, S. Kumar, and S.-F. Chang, "On the difficulty of nearest neighbor search," in *Proceedings of the 29th International Conference on Machine Learning (ICML)*, 2012.

[6] M. Aumüller and M. Ceccarello, "The role of local intrinsic dimensionality in benchmarking nearest neighbor search," in *Similarity Search and Applications (SISAP)*, 2019, pp. 113–127.

[7] Z. Wang, Q. Wang, X. Cheng, P. Wang, T. Palpanas, and W. Wang, "Steiner-hardness: A query hardness measure for graph-based ANN indexes," *Proceedings of the VLDB Endowment*, vol. 17, no. 13, 2024, arXiv:2408.13899.

[8] M. Wang, X. Xu, Q. Yue, and Y. Wang, "A comprehensive survey and experimental comparison of graph-based approximate nearest neighbor search," *Proc. VLDB Endow.*, vol. 14, no. 11, pp. 1964–1978, 2021.

[9] W. Li, Y. Zhang, Y. Sun, W. Wang, M. Li, W. Zhang, and X. Lin, "Approximate nearest neighbor search on high dimensional data — experiments, analyses, and improvement," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 8, pp. 1475–1488, 2020.

[10] M. Aumüller, E. Bernhardsson, and A. Faithfull, "ANN-Benchmarks: A benchmarking tool for approximate nearest neighbor algorithms," *Information Systems*, vol. 87, 2020.

[11] E. Chávez, G. Navarro, R. Baeza-Yates, and J. L. Marroquín, "Searching in metric spaces," *ACM Comput. Surv.*, vol. 33, no. 3, pp. 273–321, 2001.

[12] J. Gao and C. Long, "High-dimensional approximate nearest neighbor search: with reliable and efficient distance comparison operations," *Proc. ACM Manag. Data*, vol. 1, no. 2, pp. 1–27, 2023.

[13] S. Jaiswal, R. Krishnaswamy, A. Garg, H. V. Simhadri, and S. Agrawal, "OOD-DiskANN: Efficient and scalable graph ANNS for out-of-distribution queries," 2022.

[14] M. Chen, K. Zhang, Z. He, Y. Jing, and X. S. Wang, "RoarGraph: A projected bipartite graph for efficient cross-modal approximate nearest neighbor search," *Proc. VLDB Endow.*, vol. 17, no. 11, pp. 2735–2749, 2024.

[15] M. Radovanović, A. Nanopoulos, and M. Ivanović, "Hubs in space: Popular nearest neighbors in high-dimensional data," *Journal of Machine Learning Research*, vol. 11, pp. 2487–2531, 2010.

[16] R. E. Barlow, D. J. Bartholomew, J. M. Bremner, and H. D. Brunk, *Statistical Inference under Order Restrictions*. Wiley, 1972.

[17] H. V. Simhadri *et al.*, "Results of the NeurIPS'21 challenge on billion-scale approximate nearest neighbor search," in *Proceedings of the NeurIPS 2021 Competitions and Demonstrations Track*, ser. PMLR, vol. 176, 2022, pp. 177–189.

[18] E. Jääsaari, V. Hyvönen, M. Ceccarello, T. Roos, and M. Aumüller, "VIBE: Vector index benchmark for embeddings," 2025, arXiv:2505.17810.

[19] A. Ploshkin, V. Tytskiy, A. Pismenny, V. Baikalov, E. Taychinov, A. Permiakov, D. Burlakov, E. Krofto, and N. Savushkin, "Yambda-5B – a large-scale multi-modal dataset for ranking and retrieval," 2025, arXiv:2505.22238.